If X takes only a finite number of values $x_1, x_2, x_3, \ldots x_n$, where $x_1 < x_2 < x_3 <, \cdots < x_n$, then the cumulative distribution function is given by

$$F(x) = \begin{cases} 0, & -\infty < x < x_1 \\ f(x_1), & x_1 \le x < x_2 \\ f(x_1) + f(x_2), & x_2 \le x < x_3 \\ f(x_1) + f(x_2) + f(x_3), & x_3 \le x < x_4 \\ \vdots & \vdots \\ f(x_1) + f(x_2) + \ldots + f(x_n), & x_n \le x < \infty \end{cases}$$

For a discrete random variable X, the cumulative distribution function satisfies the following properties.

(i) $0 \le F(x) \le 1$, for all $x \in \mathbb{R}$.

(ii) $F(x)$ is real valued non-decreasing function $\left(x < y, \text{ then } F(x) \le F(y)\right)$.

(iii) $F(x)$ is right continuous function $\left(\lim_{x \to a+} F(x) = F(a)\right)$.

(iv) $\lim_{x \to -\infty} F(x) = F(-\infty) = 0$.

(v) $\lim_{x \to +\infty} F(x) = F(+\infty) = 1$.

(vi) $P(x_1 < X \le x_2) = F(x_2) - F(x_1)$.

(vii) $P(X > x) = 1 - P(X \le x) = 1 - F(x)$.

(viii) $P(X = x_k) = F(x_k) - F(x_k^-)$.

Note

Some authors use left continuity in the definition of a cumulative distribution function $F(x)$, instead of right continuity.

11.3.4 Cumulative Distribution Function from Probability Mass function

Both the probability mass function and the cumulative distribution function of a discrete random variable X contain all the probabilistic information of X. The *probability distribution* of X is determined by either of them. In fact, the distribution function F of a discrete random variable X can be expressed in terms of the probability mass function $f(x)$ of X and vice versa.

Example 11.7

If the probability mass function $f(x)$ of a random variable X is

x	1	2	3	4
$f(x)$	$\frac{1}{12}$	$\frac{5}{12}$	$\frac{5}{12}$	$\frac{1}{12}$

find (i) its cumulative distribution function, hence find (ii) $P(X \le 3)$ and, (iii) $P(X \ge 2)$

Solution

(i) By definition the cumulative distribution function for discrete random variable is

$$F(x) = P(X \le x) = \sum_{x_i \le x} P(X = x_i)$$

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